AI/ML-DRIVEN IOT USE CASES IN ELECTRICITY UTILITIES

Transforming Smart Metering & Grid Operations via Advanced Analytics

This white paper explores the convergence of IoT-enabled Advanced Metering Infrastructure (AMI) and custom machine learning forecasting engines. By leveraging standard grid telemetry, the reference framework deploys low-latency anomaly detection, predictive load schedulers, and network optimization algorithms localized to standard Indian Electricity Grid Code regulations.

Reference Document Details:
Context: Smart Meter National Programme (SMNP) & Advanced Infrastructure Rollouts
Technical Baseline: Smart Grid Sync Simulation Engine
Security Classification: LNT Construction Internal Use

1. Executive Summary & Industry Challenges

Electricity distribution utilities worldwide are undergoing a fundamental transformation driven by the convergence of IoT, Advanced Metering Infrastructure (AMI), and AI/ML. What began as large-scale smart meter rollouts is now evolving into data-centric grid intelligence. Utilities extract predictive, prescriptive, and real-time insights from massive streams of meter and network data.

Core Operational Challenges Addressed:

- **Non-Technical Losses (NTL):** Electricity theft, line tampering, and unmetered connection taps causing severe financial deficits (AT&C losses) in distribution grids.
- **Delayed Visibility:** Operational inefficiencies driven by manual meter readings and delayed grid state visibility, hindering outage detection.
- **High O&M Costs:** Reactive fault management and high overheads from dispatching maintenance crews after outages occur instead of predicting them.
- **Billing Inaccuracies:** Gaps in customer experience and trust driven by delayed bills, estimated readings, and slow resolution cycles.

2. AI/ML Analytics as the Operational Value Multiplier

By applying machine learning algorithms to historical smart meter telemetry, utilities can pre-plan generation and battery scheduling, prevent line congestion, and optimize load profiles. We validated three custom regressors on historical grid intervals with weather variables:

Forecasting Model	RMSE (MW)	MAE (MW)	R-Squared (Accuracy Rating)
Random Forest Regressor	5.89 MW	4.38 MW	0.953 (95.3% Variance Explained)
Gradient Boosting Regressor	5.99 MW	4.50 MW	0.952 (95.2% Variance Explained)
LSTM (PureMLP Neural Net)	11.33 MW	9.03 MW	0.828 (82.8% Variance Explained)

Solar Forecasting Integration

Using our custom NumPy XGBoost decision tree, solar generation output is predicted with a Root Mean Squared Error (RMSE) of 1.95 MW and an R-squared score of 0.997. This allows grid controllers to accurately forecast renewable generation patterns and pre-schedule battery charging.

3. Anomaly Detection & Regulatory Compliance

A. Theft & Tampering Detection (Anomaly Rules)

We integrated an explicit theft anomaly detection rule. If the smart meter records consumption dropping below 35% of the expected hourly base demand (without pricing incentives like off-peak rebates), or if the operator activates the 'Tampering / Theft' hotkey, the system flags a 'theft' anomaly. This triggers visual red pulsing alerts on dispatch paths and logs critical tamper alarms in the SQLite database, matching real-world bypass and line tapping theft alert behaviors.

B. Indian Grid Frequency Compliance (IEGC 50 Hz)

Grid frequency was localized to the Indian Electricity Grid Code standard base of 50.00 Hz, with safe operating fluctuations clamped between 49.85 Hz and 50.15 Hz. Real-time deviations are registered and recorded in database logs, allowing statistical analysis of grid stability factors.

C. Time-of-Day (ToD) Tariff & slab pricing economics

Dynamic pricing models Time-of-Day schedules: Night off-peak rebates (Rs. 1.50/unit) and Peak surcharges (Rs. 1.50 to Rs. 2.50/unit) coupled with Indian state slab-billing rates (Rs. 4.50 to Rs. 15.00/unit). This drives automated demand response, shaving consumer load by 15% during peak periods.

4. 4-Layer IoT reference Architecture

The smart grid platform serves as a pilot model for the 4-layer utility IoT architecture:

Layer 1: Physical Edge Layer

Advanced Metering Infrastructure (AMI) smart meters, battery storage controller loops, and environmental weather sensors.

Layer 2: Communication Layer

CORS-enabled RESTful APIs, asynchronous AJAX request handlers, and JSON network data exchanges.

Layer 3: Analytics & Application Layer

A Python-based multithreaded web handler, an SQLite relational logging database with a rolling 24-hour auto-pruning window (144 logs), and custom-written NumPy machine learning regressors.

Layer 4: Presentation Layer

A glassmorphic browser dashboard utilizing inline SVG drawing engines to render dispatch lines, forecasting curves, and pulsing anomaly markers.